

KRISH PATEL

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EDUCATION

University of Illinois at Urbana-Champaign

May 2025

Bachelor's in Computer Engineering

GPA: 3.75/4.00

Relevant Courses: Computer Architecture, Machine Learning, Operating system, Data Science, Probability, Artificial intelligence, Algorithm, Digital system, Database, VLSI Design

SKILLS

Programming language: Python, Embedded C, C/C++, x86 (Assembly), RISC, System Verilog, MySQL

Skills: FPGA development, Natural Language Processing, Processor Design, Data analysis, Machine learning

Tools: PyTorch, TensorFlow, Verdi, Linux, Github, SQL, gdb, MongoDB, Synopsys, Quartus, CI/CD, Docker, Qemu, Vivado, Regex, GCP, Xilinx, cadence

PROJECTS

Pipeline Out-of-Order RISC-V Processor, *Developer*

krish-patel.com/ooo

- Designed an out-of-order processor from scratch compliant with the RISC-V ISA, featuring a 4-way associative cache, D and I Cache with PLRU, reservation stations, and a reorder buffer
- Improved performance metrics by 40% through age-ordered issuing, early branch recovery, and GShare branch prediction, achieving an IPC of 0.7 on CoreMark
- Ranked in the top 15 among peers via benchmarking and testing using coverage points and random testbenches while maintaining a compact area of 200k nm^2 , enhancing hardware scalability

Unix-Like Operating System, *Developer*

krish-patel.com/os

- Developed a real-time OS kernel from scratch in x86 assembly and C, with a focus on memory mapping and virtualization. Led core tasks, including GDT/IDT setup, Paging, driver development, and scheduling
- Engineered essential device drivers for Keyboard, terminal, and RTC setup. Optimized system calls and PCBs with assembly linkage, concurrently implementing a read-only file system from scratch
- Synchronized system calls to implement round robin scheduling, facilitating a multi-terminal setup for up to 6 processes. Scripted unit tests for each function and debugged at the kernel level using gdb

Mortal Kombat Arcade Game on FPGA, *Developer*

krish-patel.com/mortal-kombat

- Replicated Mortal Kombat on an FPGA as a SoC using system verilog and Xilinx with 213 KiB of On-Chip Memory, featuring multiplayer functionality, health point tracker, and a timer for seamless gameplay
- Implemented the SPI protocol for communication between the MicroBlaze processor and the MAX3421E USB Host Controller, enabling efficient data exchange for game controls via a USB keyboard
- Integrated an AXI UART interface for serial communication, allowing data flow between MicroBlaze processor and peripherals, including HDMI, USB, and GPIO modules for efficient game control and display

EXPERIENCES

IBM-IL IIDAI Research, *Research Assistant*

May 2024 - present

- Optimized tensor compilers by implementing NLP-based search space reductions, enhancing multi-platform compatibility and significantly reducing AI compilation time
- Reproduced results from research projects with <2% variance, while migrating legacy codebases to an upgraded TVM framework for enhanced compatibility and efficiency in AI model execution
- Simplified complex research into posters and abstracts, fostering collaboration through effective presentations

Undergraduate Course Assistant,

January 2023 - Present

Computer Architecture (ECE411) | prev: Discrete math (CS173), Introduction to electronics (ECE110)

- Mentoring a class of over 100 students in debugging RISC-V HDL code and testbenches, while redesigning course projects to incorporate advanced verification techniques

NeoSOFT Technologies, *Intern*

June 2023 - August 2023

- Developed a resume parsing API using deep learning in NLP, trained on 1400+ resumes with 90% accuracy in extracting candidate information: education, skills, contact details, and experience
- Employed BeautifulSoup and Selenium to develop a web bot that efficiently scraped 1000+ job listings, storing them directly in a MongoDB database for precise candidate-to-job matching
- Designed a robust skill extraction script using advanced Regex patterns to extract 17,300+ skills from diverse resumes, bolstering insights into candidate qualifications for resume parsing